Triple Change Detection Network via Joint Multifrequency and Full-Scale Swin-Transformer for Remote Sensing Images

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***Abstract*— Although deep learning-based change detection (CD) methods achieve great success in remote sensing images, they still suffer from two main challenges. First, popular convolutional neural networks (CNNs) are weak in extracting discriminated features focusing on changed regions, since most methods ignore the multifrequency components of bitemporal images. Second, although existing CD methods employ the Transformer structure to capture long-range dependency for global feature representation, it is difficult for them to simultaneously take into account the long-range dependency of changed objects at various scales. To address the above issues, we propose a triple CD network (TCD-Net) via joint multifrequency and full-scale swin-transformer (FST). The proposed TCD-Net has two main advantages. First, we propose a multifrequency channel attention (MFCA) module to boost the ability of modeling the channel correlation, which can compensate for the problem of insufficient feature representation caused by only performing global average pooling (GAP). Furthermore, a joint multifrequency difference feature enhancement (JM-DFE) guiding block is proposed to improve the boundary quality and the position awareness of truly changed objects, which can effectively extract channel features of multifrequency information and thus improve the discriminative ability of features. Second, unlike Siamese-based structures, we propose an FST module as the third branch to model and aggregate the long-range dependency of multi-scale changed objects, which can alleviate the missed detections of small objects and achieve more compact changed regions effectively. Experiments on three public CD datasets exhibit that the proposed TCD-Net achieves better CD accuracy with smaller model complexity than state-of-the-art (SOTA) methods. The code is publicly available at https://github.com/RSCD-mz/TCD-Net.**

***Index Terms*— Change detection (CD), full-scale swin-transformer (FST), joint multifrequency features, pyramid decoder.**

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I. Introduction

CHANGE detection (CD) is the process of identifying changes that occurs in the same geographical location but at different times, which is one of the most popular technologies in Earth vision. In recent decades, CD of remote sensing images has become of great significance for many real-world applications, including urban management [1], environmental investigations [2], and damage assessment [3]. Specifically, the CD is a pixel-to-pixel task that uses coregistered bitemporal images as input data and predicts "where" the changes occurred.

Early CD methods detect changed regions by exploiting spectral information from remote sensing images, such as change vector analysis [4], thresholding methods [5], principal component analysis [6], clustering [7], and Markov random field [8]. However, these traditional CD methods face two main problems. First, these methods need to be developed according to the characteristics of remote sensing images, since they are mostly designed based on hand-crafted features and are thus sensitive to noise. Second, these methods often focus on the strong and abrupt changes, the sharp contrast between foreground and background. Thus, they are often unsuitable for low-contrast remote sensing images.

Nowadays, deep learning-based methods have shown the great success in CD applications [9]. For single-stream networks, the results of subtracting or concatenating bitemporal remote sensing images are fed into a single neural network to produce the prediction map of CD [10]. Later, to fully utilize the information of bitemporal remote sensing images, most networks adopt Siamese branches for feature extraction [11], [12]. In addition, to achieve better CD performance, some CD methods construct contextual relationships by using

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attention mechanisms, including squeeze-and-excitation (SE) [13], convolutional block attention module (CBAM) [14], etc. But these attention-based methods ignore multifrequency components of changed regions from bitemporal images by using global average pooling (GAP) alone. In particular, these multifrequency features play a significant role in identifying changed regions, such as building changes, vegetation changes, and other changes. Therefore, the multifrequency difference features are crucial for improving CD results of remote sensing images. Moreover, some CD methods model the long-range dependency by introducing vision transformer (ViT) [15], [16]. However, these methods are weak in detecting targets of various sizes accurately. Thus, the feature extraction at different scales in bitemporal remote sensing images still needs to be strengthened.

To address the above issues, we propose a triple CD network (TCD-Net) via joint multifrequency and full-scale swin-transformer (FST) for remote sensing images. By using the joint multifrequency difference feature enhancement (JM-DFE) guiding blocks, the proposed TCD-Net can capture the channel features of multifrequency information and enhance the discriminative ability of features effectively, which improves boundary quality and the position awareness of truly changed regions. Moreover, the FST module proposed in this article can extract multiscale changed features accurately by modeling the long-range dependency of multiscale changed objects. Meanwhile, the corresponding features of the FST module are introduced into the pyramid decoder module to achieve higher CD accuracy. In summary, the main contributions of this article are summarized as follows.

1) We design a multifrequency channel attention (MFCA) module to overcome the problem of insufficient feature representation for bitemporal remote sensing images. The MFCA module can boost the correlation between different feature channels and capture the channel features of multifrequency information. Based on the MFCA module, a JM-DFE guiding block is presented for CD to extract the channel features of multifrequency information and enhance the discriminative ability of features, which can improve the position awareness of changed information, and further identify changed regions with fine-details.
2) We propose an FST module as the third branch of the proposed network architecture. The FST module extracts features of both largely and little changed objects from bitemporal remote sensing images. Unlike pure swin-transformers (STs), it uses a new method, namely the full-scale method, to model long-range dependency of multiscale changed objects, which can alleviate the missed detections of small objects and achieve more compact changed regions effectively.
3) The proposed TCD-Net achieves the best CD performance on three CD datasets, LEVIR-CD, WHU-CD, and SYSU-CD, with the highest *F1* values of 92.5%, 80.2%, and 91.7%, respectively. Moreover, it also provides a smaller model size of 43.54M with fewer parameters of 11.13M.

The remainder of this article is organized as follows. The related work provides an overall review of the CD methods as shown in Section II. Section III provides a detailed description of the proposed method. Then, the experiments and results are presented in Section IV. Section V discusses the effectiveness of the proposed modules. Finally, Section VI concludes our work.

II. Related Work

A. Deep Learning-Based Models

Deep learning-based CD methods have shown excellent performance over hand-crafted features and demonstrated strong abilities of feature extraction in CD applications [17]. Prior studies have shown that convolutional neural networks (CNNs) and fully convolutional networks (FCNs) are among the most popular deep learning structures. They are often used as backbones or feature extractors for predicting the decision map due to their good generalization ability. Based on these backbones, some single-stream structures are employed to achieve CD tasks [18], [19]. These methods directly apply original difference images to corresponding models to obtain changed regions, so they can be considered as single-stream structures. Although these single-stream methods have achieved improvements on CD tasks compared to conventional methods, they ignore the original features of the bitemporal remote sensing images, since lots of detail information is missed in difference images.

To address the above problems, some Siamese-stream methods are designed based on the significance and requirements of CD [20], [21]. Typically, these Siamese-stream CD methods use two branches of subnetworks with the same structures to extract pre- and post-temporal features, separately. To learn deep bitemporal features, the two subnetworks of these Siamese-stream CD methods are trained simultaneously, which makes the Siamese structures popular in CD tasks. Moreover, these Siamese-based CD methods mainly consist of genuine-Siamese structures [22], [23] and pseudo-Siamese structures [24], [25]. Their distinction is that the former group uses a manner of sharing weights, while the latter group uses a manner of nonsharing weights. Despite the fact that these pseudo-Siamese structures increase the model complexity, they increase the flexibility of networks and are favored in CD tasks due to their effectiveness for feature extraction of bitemporal images.

Recent deep learning-based CD methods can be coarsely categorized into three types: feature fusion-based methods [26], [27], multiscale modeling-based methods [28], [29], [30], and attention-based methods [31], [32], [33]. Specifically, the first group focuses on feature fusion, which involves the combination of features from two branches of the same scale level by employing the operations such as difference, summation, or concatenation [34]. By integrating hierarchical feature maps and bridging the gap between different-level feature maps, these feature fusion-based methods have shown their effectiveness in improving CD accuracy [35], [36]. In the second group, multiscale strategies are particularly advantageous as they can improve the problems of missed details and low detection accuracy, and strengthen the ability of context modeling [28]. For instance, Zhang and Shi [29] constructed a feature difference by adopting the VGG16 as a pretrained

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model, which extracts spatial features at multiscale feature levels and exhibits a strong ability of feature representation. Moreover, in the third group, attention-based methods are particularly important for coping with the complexity of the changed regions and background from bitemporal remote sensing images. Some examples of these structures include [31], [32], and [33]. Notably, it is widely recognized that the original channel-based attentions utilize GAP to capture the features of each channel, where the result of GAP is proportional to the lowest frequency component of 2-D discrete cosine transform (2-D-DCT). However, these attentions lead to clear information loss.

To summarize, deep learning-based CD methods offer several advantages over traditional methods, including better performance, scalability, and end-to-end learning. Compared to single-stream networks, Siamese branches with feature-extracting strategies are better suited for CD tasks. Furthermore, these methods excel at extracting the contextual information of changes and detecting subtle variations. However, despite their effectiveness in analyzing bitemporal images, they may face challenges in capturing discriminative features in remote sensing images, resulting in blurred details.

B. Vision Transformer-Based Models

Transformer is a new type of deep learning model that has been widely applied to computer vision tasks. It was first used in natural language processing to achieve sequence-to-sequence tasks [37]. Later, it becomes popular in various computer vision tasks, including object detection [38], image classification [39], image segmentation [40], and video processing tasks [41]. Studies on transformer networks in computer vision have resulted in breakthroughs such as the detection transformer model [42], the ViT [43], and shift window transformers (ST) [44]. These works bring numerous advances in computer vision due to the strong global modeling ability of transformer networks.

Meanwhile, the transformers have attracted enormous research interest in the field of remote sensing owing to their outstanding performance. Recently, more and more works employed transformer models to achieve object detection [45], [46], semantic segmentation [47], [48], land classification [49], [50], CD [15], [16], [51], and other remote sensing tasks. These methods are making impressive progress to some extent. For instance, a transformer-based correlation learning detector [45] has been used to detect dense objects, which improves the detection accuracy for object detection of remote sensing images. In addition, to improve the overall performance, a hybrid deep CNN with transformer [48] has been designed for image semantic segmentation. Moreover, the ViT works as a teacher model to guide the student model for achieving land classification [50]. Chen et al. [51] proposed a transformer-based network for bitemporal remote sensing images, which employs a transformer encoding structure (ES) to model global contexts within the concatenated tokens, and a transformer decoding structure to refine original deep features by using the richer context tokens.

In summary, these methods offer several advantages over traditional CNNs commonly used in computer vision, including better ability of modeling long-range dependency between input and output data, and strong flexibility to variations in input data [48], which makes transformers more reliable for real-world computer vision applications. However, they ignore the long-range dependency of largely and little changed objects, leading to poor detection on fine edges and small-sized objects.

III. Methods

A. Overview

This section provides a comprehensive explanation for the proposed approach. The proposed TCD-Net follows a standard encoder-decoder structure. The overall model structure is illustrated in Fig. 1 and a brief overview of the TCD-Net is given as follows.

The proposed TCD-Net is composed of two stages: the encoding stage and the decoding stage. In the encoding stage, it uses triple branches to achieve feature extraction. Specifically, in the Siamese branches, the TCD-Net utilizes nonsharing weights Siamese convolutional layers (Conv-Layer) with the JM-DFE guiding blocks, which takes bitemporal images as inputs in Siamese branches. The JM-DFE guiding block aims to improve the position awareness of truly changed objects and refine the fine-details and boundaries. Meanwhile, there are ten Swin-blocks in the third branch of the FST module. Each block accepts the difference features from the same scale level of the Siamese branches, which aims to alleviate the missed detections of small objects. In the decoding stage, full-scale difference features are imported to the pyramid decoder module to generate more complete and compact changed regions. Besides, all symbols and shapes in this structure are explained in Fig. 1 (top). Moreover, the S^L denotes the corresponding features at the Lth layer, and the $X^{L,j}$ refers to the output of the Lth layer in the pyramid decoder module.

In this section, we begin by providing an elaborate explanation of the MFCA module. Subsequently, based on the MFCA module, we propose the JM-DFE guiding block for exploring how it can be used to improve changed regions. Additionally, we present the FST with the pyramid decoder module, which can achieve better CD results by leveraging FST blocks. Furthermore, according to the above illustration, the detailed design of triple branches in our TCD-Net structure is shown in Tables I and II, where the size of bitemporal images is $H \times W$, convolutional layers (Conv-Layer) denoted by stack depthwise separable convolutional layers (Depth-conv 3×3), "dim 96" denotes a linear layer with an output dimension of 96, and "win. size 7×7" indicates a multihead self-attention module with window size of 7×7, respectively.

B. Joint Multifrequency Difference Feature Enhancement Guiding Blocks

1) Global Average Pooling Alone Is Insufficient: With the advent of deep learning techniques, various attention mechanisms have been successfully introduced into CNNs [31], [32]. In these attention mechanisms, GAP is commonly

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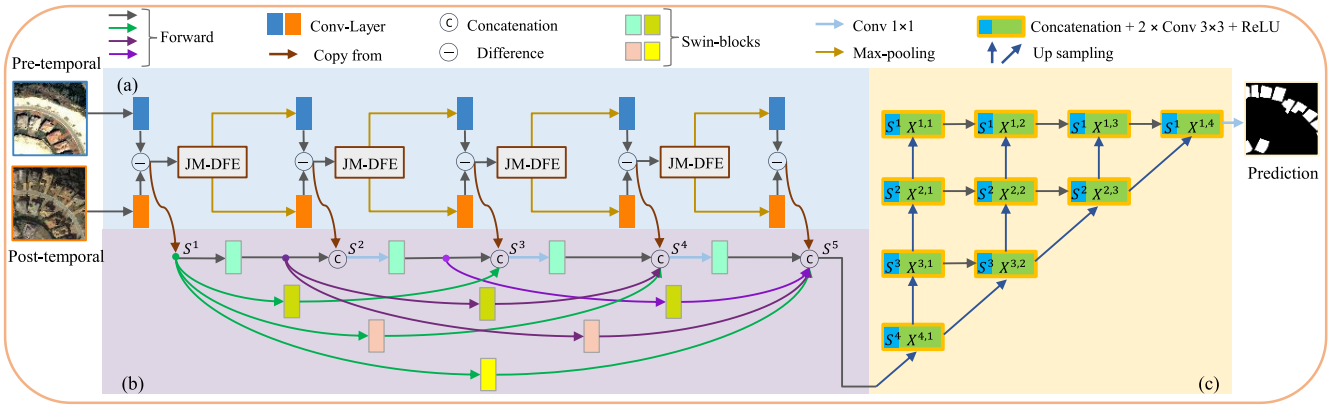


Fig. 1. Overall structure of the TCD-Net. (a) Siamese branches with JM-DFE guiding block. (b) Third branch of FST module. (c) Pyramid decoder module.

TABLE I
DETAILED STRUCTURE SPECIFICATIONS OF THE SIAMESE BRANCHES

Stage	Stage output($C \times H \times W$)	Layer-operators
1	64×H×W	2×Dept-Conv 3×3 JM-DFE (64) guiding block
2	128×$\frac{H}{2}$×$\frac{W}{2}$	Max-pool 2×2, 2×Dept-Conv 3×3 JM-DFE (128) guiding block
3	256×$\frac{H}{4}$×$\frac{W}{4}$	Max-pool 2×2, 2×Dept-Conv 3×3 JM-DFE (256) guiding block
4	512×$\frac{H}{8}$×$\frac{W}{8}$	Max-pool 2×2, 2×Dept-Conv 3×3 JM-DFE (512) guiding block
5	512×$\frac{H}{16}$×$\frac{W}{16}$	Max-pool 2×2, 2×Dept-Conv 3×3 JM-DFE (512) guiding block
6	256×$\frac{H}{8}$×$\frac{W}{8}$	Trans-Conv 2×2 2×Dept-Conv 3×3
7	128×$\frac{H}{4}$×$\frac{W}{4}$	Trans-Conv 2×2 2×Dept-Conv 3×3
8	64×$\frac{H}{2}$×$\frac{W}{2}$	Trans-Conv 2×2 2×Dept-Conv 3×3
9	64×H×W	Trans-Conv 2×2 2×Dept-Conv 3×3
Classifier	1×H×W	Conv 1×1

TABLE II
DETAILED STRUCTURE SPECIFICATIONS OF THE THIRD BRANCH

Down. rate (Output size)	Node	1	2	3	4
1× ($H \times W$)	1	—	—	—	—
2× ($\frac{H}{2} \times \frac{W}{2}$)	2	[win. size 7×7 dim 96, head 4]	—	—	—
4× ($\frac{H}{4} \times \frac{W}{4}$)	3	[win. size 7×7 dim 96, head 4]	[win. size 7×7 dim 96, head 4]	—	—
8× ($\frac{H}{8} \times \frac{W}{8}$)	4	[win. size 7×7 dim 128, head 4]	[win. size 7×7 dim 128, head 4]	[win. size 7×7 dim 128, head 4]	—
16× ($\frac{H}{16} \times \frac{W}{16}$)	5	[win. size 7×7 dim 192, head 4]	[win. size 7×7 dim 192, head 4]	[win. size 7×7 dim 192, head 4]	[win. size 7×7 dim 192, head 4]

employed as a standard down-sampling operation. Among various attention mechanisms, the SE [13] block is a popular attention mechanism that improves the representation ability of CNNs by modeling the interdependency of different channels. Specifically, the "Squeezing" step adopts GAP to generate channelwise descriptors, while the "Excitation" step aims to capture channelwise dependency and produce an attention vector. However, we argue that the use of GAP in the SE channel attention alone only preserves the lowest frequency component information. We prove this statement in the following analysis.

We suppose that $\boldsymbol{F} \in \mathbb{R}^{C \times H \times W}$ is the feature tensor in the encoding stage of networks, C is the number of channels, and $H \times W$ is the size of the current feature tensor. $D_{h,w}^{i,j}$ is used to present the basis function of 2-D-DCT, which is defined as

$$D_{h,w}^{i,j} = \cos\left(\frac{\pi h}{H}\left(i+\frac{1}{2}\right)\right)\cos\left(\frac{\pi w}{W}\left(j+\frac{1}{2}\right)\right) \tag{1}$$

where $h \in 0, 1, \ldots, H-1$, $w \in 0, 1, \ldots, W-1$. Correspondingly, to clarify the process, the result of 2-D-DCT at index (h, w) is calculated as follows:

$$\boldsymbol{f}_{\mathrm{DCT}}^{:,h,w} = \sum_{i=0}^{H-1}\sum_{j=0}^{W-1} D_{h,w}^{i,j}\boldsymbol{F}^{:,h,w} \tag{2}$$

then, the feature tensor $\boldsymbol{F}$ can be decomposed into a weighted sum of multifrequency components by using inverse DCT, which results in a decomposed feature tensor that can be presented as follows:

$$\begin{aligned}\boldsymbol{F}^{:,i,j} &= \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} \boldsymbol{f}_{\mathrm{DCT}}^{:,h,w} D_{h,w}^{i,j} \\ &= \boldsymbol{f}_{\mathrm{DCT}}^{:,0,0} D_{0,0}^{i,j} + \boldsymbol{f}_{\mathrm{DCT}}^{:,0,1} D_{0,1}^{i,j} + \cdots + \boldsymbol{f}_{\mathrm{DCT}}^{:,H-1,W-1} D_{H-1,W-1}^{i,j}\end{aligned} \tag{3}$$

where $\boldsymbol{f}_{\mathrm{DCT}}^{:,0,0} D_{0,0}^{i,j}$ denotes the lowest frequency component of 2-D-DCT. This component corresponds to the result by using GAP operation and represents fundamental feature information. It is important to note that feature information is distributed across various frequency components. Therefore, the use of GAP alone can lead to insufficient feature extraction and potential loss of crucial information.

2) Multifrequency Channel Attention Module: Inspired by the above analysis, the MFCA module as shown in Fig. 2(a) is proposed to emphasize the important information by reweighting multifrequency bands and channels. For the MFCA module, we select k frequency components that are nearest the lowest frequency component. Here, we set $k = 16$ in the MFCA module to implement the CD task. Specifically, from formula (2), the weight of the mth frequency spectrum denoted

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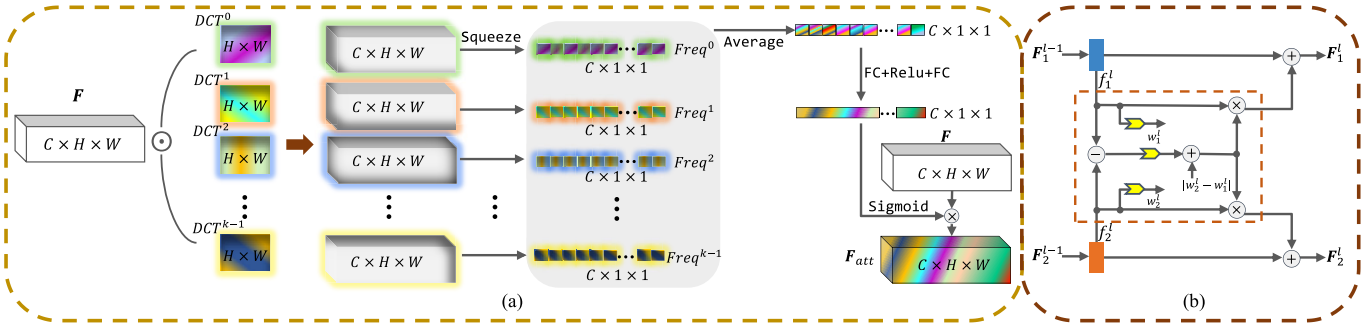


Fig. 2. (a) MFCA module. (b) JM-DFE guiding block. In the JM-DFE guiding block, the blue and orange represent the Siamese convolutional layers, the yellow denotes the MFCA module, "−" refers to the difference operation, "×" denotes channelwise multiplication, and "+" presents elementwise summation, respectively.

by Freq^m can be written as

$$\begin{aligned}&\text{Freq}^m\\&=\text{Squeeze}\left(\boldsymbol{F}\odot DCT^m\right)\\&=\text{Squeeze}\left(\sum_{i=0}^{H-1}\sum_{j=0}^{W-1}D_m^{i,j}\odot\boldsymbol{F}\right),\quad \text{s.t. } m\in\{0,1,\ldots,k-1\}\end{aligned}\tag{4}$$

where $\text{Freq}^m\in\mathbb{R}^{C\times1\times1}$, the operation of "Squeeze" aims to generate channelwise descriptors, $\text{DCT}^m(\cdot)$ denotes the symbol for calculating the mth frequency component, $D_m^{i,j}$ represents the weight information of the mth frequency component at index (h, w), and $\odot$ refers to elementwise multiplication.

Then, the weight result of the multifrequency spectrum denoted by $\text{att}\in\mathbb{R}^{C\times1\times1}$ is calculated as follows:

$$\text{att}=\text{Average}\left(\sum_{m=0}^{k-1}\text{Freq}^m\right),\quad \text{s.t. } m\in\{0,1,\ldots,k-1\}\tag{5}$$

where the operation of "Average" is referred to performing weights fusing.

After that, the final MFCA denoted by $\text{MF}_{\text{att}}\in\mathbb{R}^{C\times1\times1}$ is calculated as follows:

$$\text{MF}_{\text{att}}=S(\text{fc}(\delta(\text{fc}(\text{att}))))\tag{6}$$

where fc refers to the fully connected layer, δ stands for the ReLU function, and $s(\cdot)$ represents the sigmoid function.

Finally, the output of the MFCA module can be written as

$$\boldsymbol{F}_{\text{att}}=\boldsymbol{F}\times\text{MF}_{\text{att}}\tag{7}$$

where $\boldsymbol{F}_{\text{att}}\in\mathbb{R}^{C\times H\times W}$, and "×" denotes channelwise multiplication. In this way, the MFCA module can effectively capture the correlation between different channels, since it obtains more crucial information from other frequency components and is more representative for enhancing the discriminative ability of features.

3) Joint Multifrequency Difference Feature Enhancement Guiding Blocks: The attention mechanisms have demonstrated impressive results in the field of computer vision, and they have become popular approaches to concentrate on differential parts for CD tasks [52]. Here, we build a novel feature enhancement guiding block based on the MFCA module, which aims to improve the discriminated ability of difference features. We refer to this block as the JM-DFE guiding block, as depicted in Fig. 2(b). The JM-DFE guiding block comprises two subbranches and a joint MFCA module in the encoding stage.

Let us consider two output feature tensors $\boldsymbol{F}_1^{L-1}$, $\boldsymbol{F}_2^{L-1}\in\mathbb{R}^{C\times H\times W}$ from previous layers as two inputs of the current Lth layer. The MFCA module is then executed on the two feature tensors, as well as their difference feature tensor in the current Lth layer, respectively. To simplify the inference process, we employ "MF(·)" to represent the process of formulas (4)–(6). As a result, we obtain the attention weights w_1^L, w_2^L, and w_d^L corresponding to $\boldsymbol{f}_1^L$, $\boldsymbol{f}_2^L\in\mathbb{R}^{C'\times H'\times W'}$ and $\boldsymbol{f}_d^L$ as follows:

$$\begin{aligned}\boldsymbol{w}_1{}^L&=\text{MF}(\boldsymbol{f}_1^L),\quad \boldsymbol{f}_1^L=\text{Conv}(\boldsymbol{F}_1^{L-1})\\\boldsymbol{w}_2^L&=\text{MF}(\boldsymbol{f}_2^L),\quad \boldsymbol{f}_2^L=\text{Conv}(\boldsymbol{F}_2^{L-1})\\\boldsymbol{w}_d^L&=\text{MF}(\boldsymbol{f}_d^L),\quad \boldsymbol{f}_d^L=|\boldsymbol{f}_2^L-\boldsymbol{f}_1^L|\end{aligned}\tag{8}$$

where $\boldsymbol{w}_1^L$, $\boldsymbol{w}_2^L$, $\boldsymbol{w}_d^L\in\mathbb{R}^{C'\times1\times1}$, and "Conv" refers to a convolutional block that consists of two convolutional layers. Each convolutional operation includes corresponding filters of size 3×3 followed by a batch normalization layer and a ReLU function.

Based on the above descriptions, the output of the JM-DFE guiding block at the Lth layer is calculated as follows:

$$\begin{aligned}\boldsymbol{F}_1^L&=\boldsymbol{f}_1^L+\boldsymbol{f}_1^L\times\left(\boldsymbol{w}_d^L+|\boldsymbol{w}_2^L-\boldsymbol{w}_1^L|\right)\\\boldsymbol{F}_2^L&=\boldsymbol{f}_2^L+\boldsymbol{f}_2^L\times\left(\boldsymbol{w}_d^L+|\boldsymbol{w}_2^L-\boldsymbol{w}_1^L|\right)\end{aligned}\tag{9}$$

where "×" denotes channelwise multiplication, "+" presents elementwise summation, and $\boldsymbol{F}_1^L$, $\boldsymbol{F}_2^L\in\mathbb{R}^{C'\times H'\times W'}$.

In this way, the JM-DFE guiding block is employed to effectively improve the position awareness of changed information and identify truly changed regions. Here, some attention maps are presented in Fig. 3. From Fig. 3(c)–(e), it can be seen that the JM-DFE guiding blocks can help the network to focus on changed features and preserve more discriminative features. This reveals that the JM-DFE guiding blocks can enhance the feature representation of significant regions and further enable our TCD-Net to capture intricate details and subtle changes, leading to more precise and reliable CD results. Consequently, the utilization of the JM-DFE guiding blocks contributes to the overall performance enhancement of the proposed method in CD tasks.

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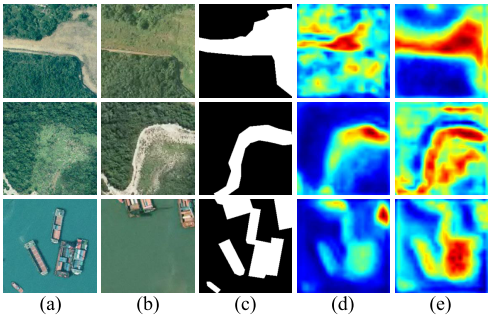


Fig. 3. Visualization of some attention maps. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. (d) Siamese convolutional ES. (e) JM-DFE guiding block. Red denotes higher attention values and blue denotes lower values.

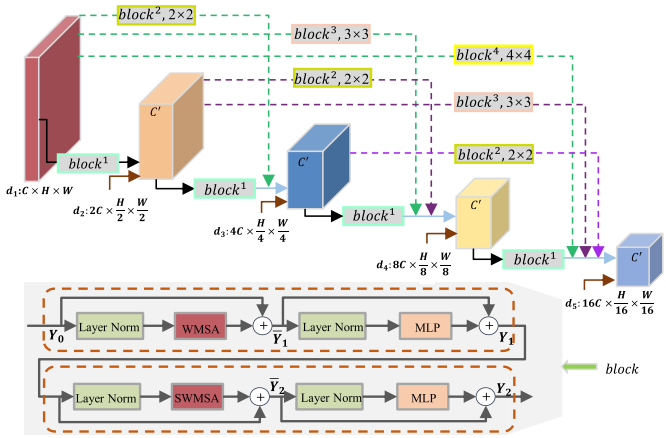


Fig. 4. FST module.

C. Full-Scale Swin-Transformer With Pyramid Decoder Module

1) Full-Scale Swin-Transformer Module: For CD tasks, Siamese-stream networks typically extract features by convolutional blocks of CNNs. However, current CD methods are difficult to consider multiscale changed objects, ignoring the long-range dependency of largely or little changed objects. Thus, it is necessary to utilize a multiscale strategy to achieve satisfactory CD results. It is worth mentioning that a small version of the ST [44] is proposed as a vision backbone due to its excellent trade-off between efficiency and accuracy. Inspired by the vision of the ST, we attempt to use the ST block as the basic unit to build an FST module and learn hierarchical representations from full-scale difference features.

As shown in Fig. 4, the proposed FST module has a hierarchical transformer structure that consists of ten basic ST blocks. Here, we provide a formal description of the block as follows:

$$\begin{aligned}\overline{\boldsymbol{Y}}_1 &= \mathrm{WMSA}(\mathrm{LN}(\boldsymbol{Y}_0)) + \boldsymbol{Y}_0\\ \boldsymbol{Y}_1 &= \mathrm{MLP}\big(\mathrm{LN}\big(\overline{\boldsymbol{Y}}_1\big)\big) + \overline{\boldsymbol{Y}}_1\\ \overline{\boldsymbol{Y}}_2 &= \mathrm{SWMSA}(\mathrm{LN}(\boldsymbol{Y}_1)) + \boldsymbol{Y}_1\\ \boldsymbol{Y}_2 &= \mathrm{MLP}\big(\mathrm{LN}\big(\overline{\boldsymbol{Y}}_2\big)\big) + \overline{\boldsymbol{Y}}_2\end{aligned} \tag{10}$$

where $\boldsymbol{Y}_0$ represents the input tensor of each block, and "LN" refers to the operation of layer normalization. "MLP" denotes the multilayer perceptron layer, $\overline{\boldsymbol{Y}}_1$ and $\overline{\boldsymbol{Y}}_2$ denote the output tensor of the window-based multihead self-attention (WMSA) module and shifted WMSA (SWMSA) module [42]. Additionally, $\boldsymbol{Y}_1$ and $\boldsymbol{Y}_2$ represent the middle output tensor and output tensor of the ST block, respectively.

Based on the Swin-block units, we build an FST module to improve the feature representation of multiscale changed objects. Specifically, the D^L represents the difference feature tensor of the Lth layer, and is introduced to the corresponding Swin-block. Then, the output of the Lth layer block denoted by S^L is calculated as

$$S^L = \begin{cases} D^L, & L = 1\\ \left[D^L, \left(\mathrm{block}^k\left(S^{L-k}\right)\right)_{k=1}^{L-1}\right], & L > 1\end{cases} \tag{11}$$

where in the above formula, s.t. $L \in \{1, 2, 3, 4, 5\}$, the symbol of $[\cdot]$ refers to the concatenation layer, $(\cdot)_{k=1}^{L-1}$ denotes total L input feature tensors for the corresponding node, block^k denotes the Swin-block, and its corresponding patch size is $k \times k$.

Basically, the $(L+1)$th layer receives the output of the Lth layer, and the output (S^L) of each Swin-block has a dimensionality of $(H/2^L) \times (W/2^L) \times C_L$, where H and W denote the height and width of the input tensor, and C_L is the number of channels in the current Swin-block. By doing so, the proposed FST module can establish the interconnection between adjacent Swin-blocks and nonadjacent Swin-blocks. Consequently, these Swin-blocks allow the network to learn hierarchical representations from full-scale difference feature maps.

2) Pyramid Decoder Module Based on Full-Scale Swin-Transformer Features: In general, a simple up-sampling of encoding features to multiresolution can achieve CD results. However, it is insufficient to employ contextual information in recovering semantic features. To further improve the detection accuracy, we build a pyramid decoder module based on the FST module that is shown in Figs. 1(c) and 4. Each layer output of the pyramid decoder module can be defined as

$$X^{L,j} = \mathrm{Conv}\left(\left[S^L, U\left(X^{L+1,j}\right)\right]\right) \tag{12}$$

where $j \in \{1, 2, 3, 4\}$, the symbol of $[\cdot]$ denotes the concatenation layer, and $U(\cdot)$ standards for an up-sampling layer. Additionally, "Conv" refers to a convolutional block that consists of two convolutional layers, each convolutional operation includes corresponding filters of size 3×3 followed by a batch normalization layer and a ReLU function.

In this way, the pyramid decoder module with bottom-up fusion method can utilize contextual information and long-range dependency of multiscale changed objects to improve the detection accuracy of truly changed regions. Moreover, some attention maps are displayed in Fig. 5, which shows that the model can capture more rewarding features on both largely and smally changed regions as basic ST, FST and pyramid decoder modules are progressively incorporated into the Siamese convolutional ES. In summary, the FST module yields noticeable improvements in capturing rewarding features, making the model capable in detecting multiscale changed objects. Furthermore, the proposed pyramid decoder module enhances the network's ability of identifying changed

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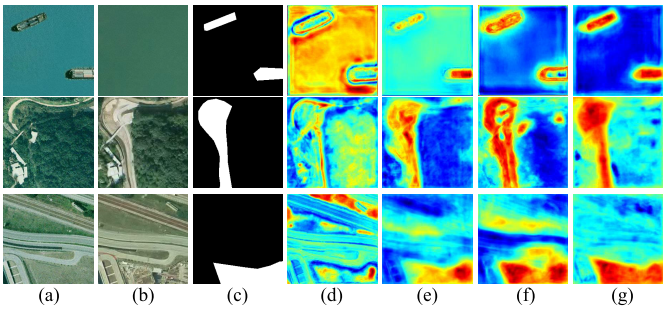


Fig. 5. Visualization of some attention maps. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. (d) Siamese convolutional ES. (e) Basic ST. (f) FST. (g) FST + pyramid decoder. Red denotes higher attention values and blue denotes lower values.

regions precisely. This demonstrates that the proposed modules can provide a better understanding of changed features for bitemporal remote sensing images.

IV. Experiments and Analysis

To evaluate and compare the proposed method, some state-of-the-art (SOTA) methods, such as FC-EF [10], FC-Siam-Di [10], FC-Siam-Conc [10], FCN-PP [11], FDCNN [29], DSIFN [53], SRCD-Net [31], Trans-CD [51], DSAMNet [33], and ChangeFormer [54] are considered as comparative methods in our experiments. Meanwhile, three public CD datasets, including the LEVIR-CD, the SYSU-CD, and the WHU-CD datasets, are first illustrated. Then, we provide a brief description of the implementation details and evaluation metrics. Moreover, the experimental results of our proposed method and the comparative methods are analyzed in detail. Finally, we conducted ablation experiments to further validate the effectiveness of our proposed module.

A. Datasets

In this article, three benchmark datasets, the LEVIR-CD, the SYSU-CD, and the WHU-CD datasets, are used to evaluate the proposed method. These datasets consist of raw bitemporal images and corresponding ground truths.

1) *LEVIR-CD [17]:* This dataset is taken between a time span of 5 and 14 years, which contains 637 pairs of bitemporal images with a size of 1024 × 1024 and a spatial resolution of 0.55 m. Their fully annotated dataset contains a total of 31 333 individual changed examples. We apply overlapping and no overlapping manners to crop the data into image patches with a size of 224 × 224, then obtain 11 083 training samples, 2880 validation samples, and 2048 testing samples, respectively.
2) *SYSU-CD [20]:*This dataset is taken between the years 2007 and 2014 in Hong Kong, which contains 20 000 pairs of orthographic aerial image patches with the original size of 256 × 256 and a spatial resolution of 0.5 m. The main types of changes in this dataset include newly built urban buildings, changes of vegetation, sea construction, and other changes. To facilitate the unified comparison of CD results, we resize the image size to 224 × 224 on the SYSU-CD dataset. Finally, this dataset provides 12 000 training samples, 4000 validation samples, and 4000 testing samples, respectively.
3) *WHU-CD [55]:* This dataset is a building CD dataset with a spatial resolution of 0.075 m. It contains one pair of bitemporal images with a size of 32507 × 15354. We first divide the bitemporal images into four smaller images without overlapping: 32 507 × 12.610/18 361 × 2.744/7634 × 2.744/6511 × 2.744. We use the first patch as the training set, the second and third patches as the validation set, and the fourth patch as the testing set. Then, we crop these data into image patches with a size of 224 × 224, obtaining 9637 training samples, 2494 validation samples, and 1600 testing samples.

B. Implementation Details

The proposed TCD-Net is implemented by PyTorch environment and trained using NVIDIA GeForce RTX 3090 GPU. In this article, the parameters setting of comparative approaches follows the original papers. For the proposed method, we train it by adopting the BCELoss. And the Adam optimizer is adopted as an optimization algorithm. Meanwhile, we trained all models with the same hyperparameters. With a learning rate of 0.0001, the batch size of all the training models was set to 16, and the number of epochs was set to 200.

C. Evaluation Metrics

To compare SOTA methods with our method, we use the intersection over union (IoU), F-score ($F1$) and the distance from the ideal position (DIP) [56] metrics as the main quantitative metrics. Above three metrics present the overall evaluation for predicting CD results. In addition, we also compare three popular metrics, including precision (Pre), recall (Rec), and overall error (OE). Specifically, the Pre denotes the ratio of detections that are really changed regions in totally detected regions. The Rec denotes the ratio of detections that are really changed regions compared to ground truths. The OE is usually used to evaluate the OE ratio of detections. These metrics are calculated as follows:

$$\begin{aligned}
\text{Pre} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\
\text{Rec} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\
\text{OE} &= \frac{\text{FP}+\text{FN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \\
\text{IoU} &= \frac{\text{TP}}{\text{FP}+\text{FN}+\text{TP}} \\
F1 &= 2 \times \frac{\text{Pre} \times \text{Rec}}{\text{Pre}+\text{Rec}} \\
\text{DIP} &= 1 - \frac{\sqrt{\sum_{n=1}^{N}(1-m_i)^2}}{\sqrt{N}}
\end{aligned} \tag{13}$$

where the above five evaluation metrics are associated with four values, i.e., TP denotes a true positive, TN denotes a true negative, FP denotes a false positive, and FN denotes a false negative, respectively. For the DIP metric, $m_i \in \{\text{Pre}, \text{Rec}, (1-\text{OE}), \text{IoU}, F1\}$. Correspondingly, N denotes the number of all evaluation metrics expect for DIP. Typically, a smaller

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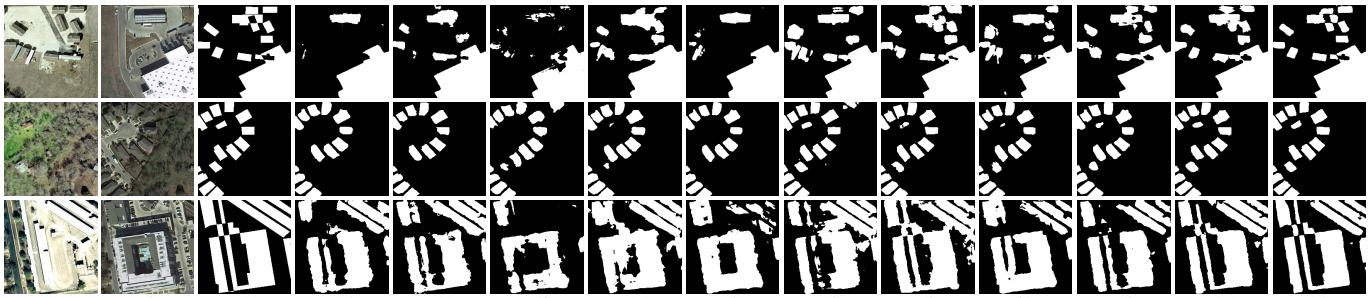


Fig. 6. Experimental results on the LEVIR-CD dataset. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. (d) FC-EF. (e) FC-Siam-Di. (f) FC-Siam-Conc. (g) FCN-PP. (h) FDCNN. (i) DFISN. (j) SRCD-Net. (k) Trans-CD. (l) DSAMNet. (m) ChangeFormer. (n) Ours.

value of OE and larger values of IoU, $F1$, and DIP mean that the better prediction CD results, and vice versa.

D. Comparisons With SOTA Methods

1) Comparison on the LEVIR-CD Dataset: Fig. 6 shows the CD results on the LEVIR-CD dataset, where Fig. 6(a)–(c) are the bitemporal images and the ground truths, respectively. From Fig. 6, it can be seen that all comparative methods roughly detect changed regions, but there are too many missed detections in the first three comparative methods as shown in Fig. 6(d)–(i). Also, DFISN generates slightly better CD results than both FCN-PP and FDCNN, since it employs a deeply supervised strategy. Meanwhile, as shown in Fig. 6(j), SRCD-Net improves the CD results using a stacked attention module. Thus, it presents a better visual effect that is close to the ground truths than the first three comparative methods. Additionally, as can be seen from Fig. 6(k)–(m), ChangeFormer maintains the internal integrity of building changes and displays a better visual effect than Trans-CD and DSAMNet. However, it is limited by the loss of low-level detail caused by direct up-sampling to full resolution. Compared to these comparative methods, multifrequency information can be leveraged effectively in the proposed TCD-Net by the introduction of our JM-DFE guiding blocks, which is able to achieve a more comprehensive representation of changed objects and a more precise positional awareness. Additionally, as shown in Fig. 6(n), the proposed TCD-Net acquires more compact changed regions than the comparative methods, and thus provides a better visual effect. The reason is that our TCD-Net combines the FST and the pyramid decoder modules to capture the correlations among changed objects of different scales. Meanwhile, our results have the fewest false detections and missed detections, while preserving the integrity of boundaries for both largely and smally changed regions.

Accordingly, it is obvious that our quantitative results on the LEVIR-CD dataset achieve better performance compared to other comparative methods, which are presented in Table III. Among the comparative methods, FC-Siam-Di provides a higher value of $F1$ among the first three comparative methods, since FC-Siam-Di considers the differences of bitemporal images in the encoding stage. Both DFISN and DSAMNet show satisfactory $F1$ values of 89.60% and 89.52%, respectively, which outperform the results provided by FCN-PP, FDCNN, and Trans-CD due to the effectiveness of deep supervision for CD. Moreover, SRCDNet and ChangeFormer achieve higher $F1$ values of 90.36% and 90.27%, respectively, since the stacked attention module is in favor of capturing changed information, and the transformer model is efficient for improving the CD accuracy. Additionally, the results reported in Table III show the superior performance of the proposed method on the LEVIR-CD dataset. Its IoU, $F1$, and DIP values are 86.45%, 92.53%, and 91.57%, respectively, which exceed the lowest result of the comparative methods by 8.29%, 5.15%, and 9.53%, and the best result of the comparative methods by 3.23%, 2.17%, and 2.23%.

TABLE III

QUANTITATIVE EVALUATION RESULTS ON THE LEVIR-CD DATASET. HIGHER VALUES OF IOU, $F1$, DIP, AND LOWER OE INDICATE GOOD CD PERFORMANCE. THE BEST VALUES ARE IN BOLD, AND ALL THE VALUES ARE IN %

Methods	Pre	Rec	OE	IoU	$F1$	DIP
FC-EF[10]	87.48	84.52	3.07	74.97	85.97	84.32
FC-Siam-Di[10]	88.70	84.38	2.70	75.96	86.49	84.90
FC-Siam-Conc[10]	86.68	79.85	3.47	72.93	83.13	82.04
FCN-PP[11]	85.81	89.01	2.73	78.16	87.38	86.10
FDCNN[29]	90.70	84.55	2.62	78.28	87.51	86.14
DFISN[53]	90.99	88.25	2.31	81.38	89.60	88.34
SRCD-Net[31]	91.75	89.01	2.09	83.22	90.36	89.34
Trans-CD[51]	90.45	86.43	2.59	78.97	88.39	86.89
DSAMNet[33]	89.27	89.76	2.40	81.20	89.52	88.26
ChangeFormer[54]	90.79	89.77	2.13	82.75	90.27	89.17
Ours	**93.55**	**91.54**	**1.67**	**86.45**	**92.53**	**91.57**

2) Comparison on the SYSU-CD Dataset: Fig. 7 shows the CD results on the SYSU-CD dataset, where Fig. 7(a)–(c) are the bitemporal images and the ground truths, respectively. In contrast to our model, it can be seen that the boundary areas of changed objects provided by comparative methods are the most damaged. Specifically, Figs. 7(d)–(f) shows that FC-EF, FC-Siam-Di, and FC-Siam-Conc generate some missed detections and false detections. Both FCN-PP and FDCNN provide CD results with some missed detections as shown in Fig. 7(g) and (h). DFISN shows fewer missed detections due to the deeply supervised strategy while it produces more false detections. In addition, compared to Fig. 7(c)–(i), Fig. 7(j) shows that SRCD-Net provides a good trade-off between missed detections and false detections due to the effectiveness of the stacked attention module for CD. Fig. 7(k) and (l)

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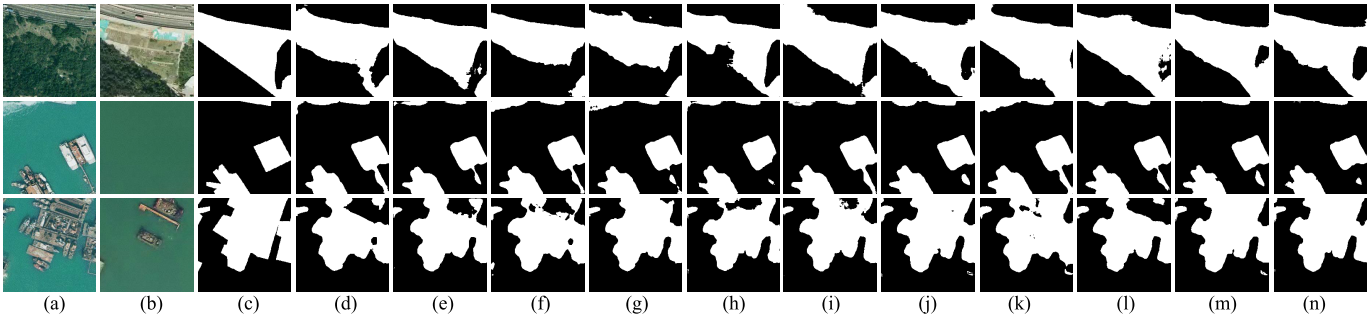


Fig. 7. Experimental results on the SYSU-CD dataset. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. (d) FC-EF. (e) FC-Siam-Di. (f) FC-Siam-Conc. (g) FCN-PP. (h) FDCNN. (i) DFISN. (j) SRCD-Net. (k) Trans-CD. (l) DSAMNet. (m) ChangeFormer. (n) Ours.

TABLE IV
QUANTITATIVE EVALUATION RESULTS ON THE SYSU-CD DATASET. HIGHER VALUES OF IOU, $F1$, DIP, AND LOWER OE INDICATE GOOD CD PERFORMANCE. THE BEST VALUES ARE IN BOLD, AND ALL THE VALUES ARE IN %

Methods	Pre	Rec	OE	IoU	$F1$	DIP
FC-EF[10]	78.63	71.50	9.62	60.39	74.90	73.32
FC-Siam-Di[10]	81.56	70.07	9.95	60.42	75.38	73.54
FC-Siam-Conc[10]	79.83	69.96	10.14	59.21	74.57	72.71
FCN-PP[11]	75.84	76.60	10.23	61.46	76.21	74.36
FDCNN[29]	78.43	74.54	10.07	61.63	76.44	74.54
DFISN[53]	82.38	73.77	9.57	63.12	77.84	75.75
SRCD-Net[31]	81.20	77.83	9.09	65.22	79.48	77.38
Trans-CD[51]	77.04	77.54	9.92	62.50	77.29	75.29
DSAMNet[33]	79.91	77.07	9.62	64.19	78.46	76.47
ChangeFormer[54]	80.01	78.52	9.14	65.02	79.26	77.20
Ours	**81.13**	**79.33**	**8.68**	**66.32**	**80.22**	**78.16**

provided by Trans-CD and DSAMNet present fewer missed detections than the first three comparative methods but refer to more false and incomplete detections. Notably, compared to other comparative methods, we can see that the results provided by ChangeFormer display a better visual effect in some boundaries of changed regions. But it detects some false changes and fails to some fine-grained changes because of the loss of details in direct up-sampling. In summary, although these comparative methods capture the main regions of changed objects, the details and boundaries of the changed regions are lost compared to Fig. 7(g)–(n). Different from these comparative methods, the TCD-Net improves the feature representation of changed objects by utilizing the JM-DFE guiding blocks. Furthermore, the FST module is employed to capture changed features more comprehensively, and further improve the compactness of changed objects and detection accuracy as shown in Fig. 7(n). Hence, the proposed TCD-Net obtains the best visual effect of changed regions with more fine-grained details on the SYSU-CD dataset.

Accordingly, Table IV reports the quantitative evaluation results on the SYSU-CD dataset. Compared to Table III, the detection accuracy of all methods on the SYSU-CD dataset is not high, but the proposed method can detect more truly changed regions. As shown in Table IV, FC-EF, FC-Siam-Di, and FC-Siam-Conc generate unsatisfactory CD results with $F1$ values of 74.90%, 75.38%, and 74.57%, respectively. FCN-PP and FDCNN achieve similar evaluation metrics on $F1$ and IoU. In addition, DFISN and DSAMNet generate higher $F1$ values of 77.84% and 77.29% than the first five comparative methods, indicating that deep supervision is useful for CD. In contrast, ChangerFormer presents better metrics by using the Siamese transformer, and SRCD-Net performs better results than the other methods due to the stacked attention module. Moreover, the proposed method achieves better results than all comparative methods. Specifically, the improvements over the lowest result of the comparative methods are 7.11%, 5.65%, and 5.45% on IoU, $F1$, and DIP, respectively. And the improvements over the best result of the comparative methods are 1.1%, 0.74%, and 0.78% on IoU, $F1$, and DIP. With respect to the best comparative method, the performance increment of our TCD-Net on the SYSU-CD dataset is more limited compared to the LEVIR-CD dataset. However, as we can see from Table IV and the corresponding visual results in Fig. 7, the proposed method is the best in all evaluation metrics and can overcome some issues, including nonsmooth boundaries and false detections.

3) Comparison on the WHU-CD Dataset: Fig. 8 shows the results on the WHU-CD dataset, where Fig. 8(a)–(c) are the bitemporal images and the ground truths, respectively. These changed targets mainly concentrate on buildings and suburban houses. It can be seen from Fig. 8 that the contrast of changed objects in bitemporal images is quite low, which may affect the accuracy of CD results. Thus, it contains rough changed boundaries and some false detections as shown in Fig. 8(d)–(h). Among the first three comparative methods, FC-Siam-Di provides a better visual effect, because FC-Siam-Di explicitly guides the network to compare the difference between the pretemporal and posttemporal images. In contrast, it can be concluded from Fig. 8(g)–(j) that SRCD-Net produces a better CD visual effect due to the stacked attention module. Meanwhile, the ChangeFormer generates better CD regions with clear boundaries as can be seen from Fig. 8(k)–(m). In Fig. 8(d)–(n), we can see that the proposed TCD-Net improves the details of changed regions, and is conducive to achieving a more complete and compact changed regions as shown in Fig. 8(n). On the whole, the proposed TCD-Net shows the best visual effect on the WHU-CD dataset.

Accordingly, the quantitative evaluation results on the WHU-CD dataset are presented in Table V. Compared to the

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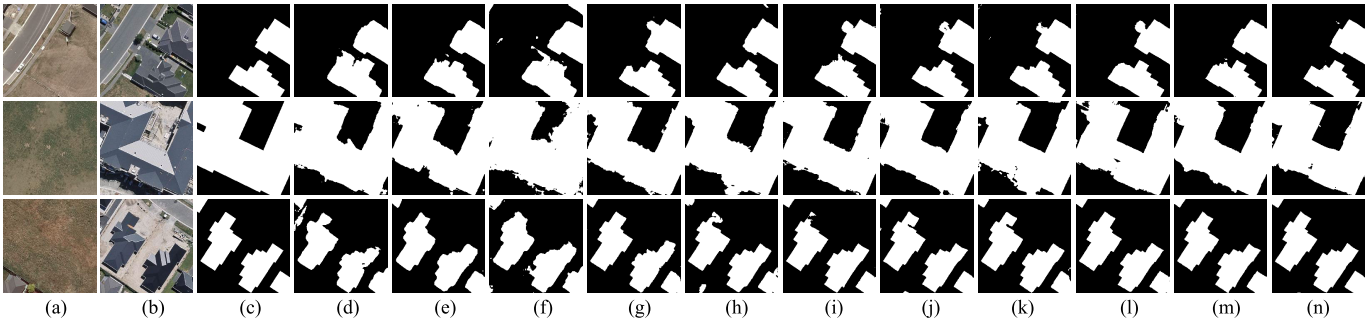


Fig. 8. Experimental results on the WHU-CD dataset. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. (d) FC-EF. (e) FC-Siam-Di. (f) FC-Siam-Conc. (g) FCN-PP. (h) FDCNN. (i) DFISN. (j) SRCD-Net. (k) Trans-CD. (l) DSAMNet. (m) ChangeFormer. (n) Ours.

TABLE V
QUANTITATIVE EVALUATION RESULTS ON THE WHU-CD DATASET. HIGHER VALUES OF IOU, $F1$, DIP, AND LOWER OE INDICATE GOOD CD PERFORMANCE. THE BEST VALUES ARE IN BOLD, AND ALL THE VALUES ARE IN %

Methods	Pre	Rec	OE	IoU	$F1$	DIP
FC-EF[10]	81.02	81.59	3.01	73.03	81.30	81.10
FC-Siam-Di[10]	88.01	82.50	2.30	77.75	85.17	84.70
FC-Siam-Conc[10]	78.81	82.71	5.06	72.27	80.71	80.43
FCN-PP[11]	89.10	85.67	1.94	80.15	87.35	86.72
FDCNN[29]	87.47	85.37	2.08	78.53	86.41	85.71
DFISN[53]	92.26	86.91	1.85	82.49	89.51	88.59
SRCD-Net[31]	92.01	88.19	1.74	83.80	90.06	89.35
Trans-CD[51]	91.29	85.57	1.69	81.95	88.34	87.78
DSAMNet[33]	90.71	87.11	1.83	82.37	88.87	82.25
ChangeFormer[54]	91.05	87.40	1.74	82.77	89.40	88.66
Ours	**93.48**	**90.06**	**1.56**	**85.52**	**91.74**	**90.82**

comparative methods, the proposed TCD-Net outperforms all comparative methods significantly on the WHU-CD dataset and achieves the highest IoU, $F1$, and DIP values of 85.52%, 91.74%, and 90.82%. As can be seen from Table V, FC-Siam-Di also achieves the best segmentation accuracy among the first three comparative methods. Moreover, SRCD-Net obtains a higher $F1$ value of 90.06% than the other comparative methods. Compared to the SRCD-Net, the proposed TCD-Net produces an extra raising of 1.72%, 1.68% and 1.47% on IoU, $F1$ and DIP values owing to better discriminative ability of features by using the JM-DFE guiding block and the effective usage of the FST-PD module. From the above analysis, the proposed TCD-Net is effective in achieving accurate CD results.

E. Ablation Experiments

In our ablation studies, we focus on exploration studies to assess the different components of the proposed method. Starting from the Siamese convolutional ES (SIAM_Conv) for CD, we incorporate the MFCA module, the JM-DFE guiding block, the FST module, and the pyramid decoder module progressively into the SIAM_Conv. The corresponding quantitative and qualitative results are summarized in Fig. 9 and Table VI, respectively. As shown in Fig. 9(d) and (e), it can

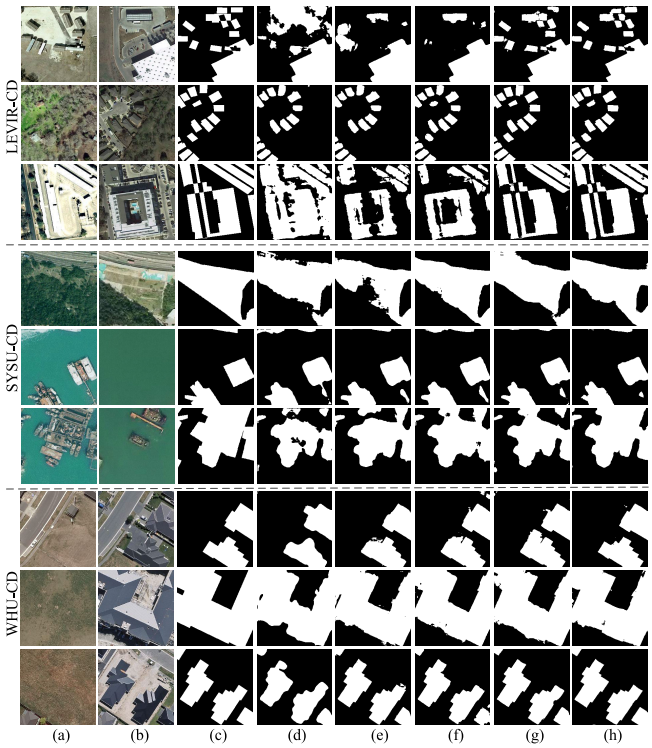


Fig. 9. Comparison of ablation experiments on the LEVIR-CD, SYSU-CD, and WHU-CD datasets. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. (d) SIAM_Conv. (e) SIAM_Conv + MFCA. (f) SIAM_Conv + JMDFE. (g) SIAM_Conv + JMDFE + FST. (h) SIAM_Conv + JMDFE + FST + pyramid decoder (Ours).

be concluded that the Siamese feature extraction methods based on the MFCA module can obtain better CD results. Fig. 9(f) shows that the JM-DFE guiding block is essential for locating the truly changed regions and improving the CD results. In addition, compared with Fig. 9(f)–(h), it shows that the results of adding the FST module capture the long-range dependency of both largely and little changed objects really well, indicating that the FST module can respond better to feature extraction of complex changes. Moreover, the results provided by adding the pyramid decoder module show the best visual effect with clear boundaries, accurate position, and high internal compactness of changed regions. It is evident from Table VI that the evaluation metrics are improved to varying degrees by adding these modules, which indicates that the proposed structure has a positive impact on CD tasks.

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TABLE VI

QUANTITATIVE EVALUATION RESULTS FOR ABLATION EXPERIMENTS ON THREE DATASETS. HIGHER VALUES OF IoU, $F1$, AND DIP INDICATE GOOD CD PERFORMANCE. THE BEST VALUES ARE IN BOLD, AND ALL THE VALUES ARE IN %

Case	Configuration	LEVIR-CD			SYSU-CD			WHU-CD		
		IoU	*F*1	*DIP*	*IoU*	*F*1	*DIP*	*IoU*	*F*1	*DIP*
I	SIAM_Conv	74.81	85.20	83.68	58.41	74.25	72.27	76.92	84.41	84.05
II	+MFCA	76.97	86.67	85.28	61.54	75.81	74.17	78.29	86.18	84.45
III	+JMDFE	78.59	87.82	86.46	61.75	76.32	74.53	79.21	86.51	85.90
IV	+JMDFE+FST	83.51	90.52	89.49	63.36	78.44	76.15	81.12	88.08	87.44
V	**+JMDFE+FST+PD**	**86.45**	**92.53**	**91.57**	**66.32**	**80.22**	**78.16**	**85.52**	**91.74**	**90.82**

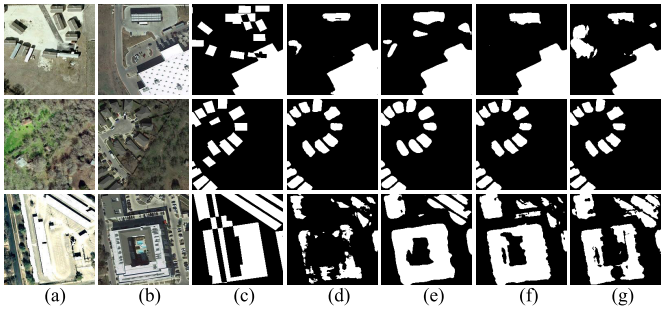


Fig. 10. Comparison of experiments on the LEVIR-CD datasets. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. The CD results of the different number of frequency components (d) $k = 1$, (e) $k = 4$, (f) $k = 8$, and (g) $k = 16$ in the JM-DFE guiding block.

V. DISCUSSION

In this section, some discussions about the sensitivity experiments of the JM-DFE guiding block, the sensitivity experiments of the pyramid decoder module based on the FST features, model performance on changed objects sizes, as well as the model complexity are presented to further illustrate the contributions of the study.

A. Discussion on the Sensitivity Experiments of the JM-DFE Guiding Block

As described in Section III-B, the JM-DFE guiding block is designed to improve position awareness of changed information and identify truly changed regions accurately. In the analysis of the JM-DFE guiding block, we tested four different values of k, representing the number of frequency components ($k = 1$, 4, 8, and 16). Their CD results and corresponding visual effects can be found in Fig. 10 and Table VII. These results exhibit that an increase in the number of multifrequency components leads to an improvement in CD results. Specifically, when $k = 16$, we observe a better visual effect as shown in Fig. 10(g) and higher accuracy of CD results with IoU and $F1$ values of 76.97% and 86.67%, respectively. As a result, we set $k = 16$ in the JM-DFE guiding block to generate CD results.

B. Discussion on the Sensitivity Experiments of the Pyramid Decoder Module

As illustrated in Section III-C, the pyramid decoder module based on FST features is presented to refine the details and boundaries of changed regions for obtaining more effective changed features. In this section, the qualitative and quantitative CD results with different layer numbers in the pyramid decoder module are shown in Fig. 12 and Table VIII, respectively. As shown in Fig. 12, the results of four pyramid decoder modules all show satisfactory visual effects. The results corresponding to the ES of the proposed TCD-Net and the bottom right structure in Fig. 11 contain more compact detections, which indicates that the pyramid decoder module based on FST features is beneficial for feature extraction in CD tasks. Simultaneously, Table VIII shows similar descriptions with correspondings visual effects. Thus, based on the above analysis, we select the bottom right structure in Fig. 11 as the decoding structure of the proposed network.

TABLE VII

QUANTITATIVE EVALUATION RESULTS FOR THE DIFFERENT FREQUENCY COMPONENTS IN THE JM-DFE GUIDING BLOCK ON THE LEVIR-CD DATASET. HIGHER VALUES OF IoU, $F1$, AND DIP INDICATE GOOD CD PERFORMANCE. THE BEST VALUES ARE IN BOLD, AND ALL THE VALUES ARE IN %

Evaluation metrics	k			
	1	4	8	**16**
IoU	75.99	76.32	76.89	**76.97**
*F*1	85.78	86.23	86.56	**86.67**
DIP	84.46	84.91	85.27	**85.28**

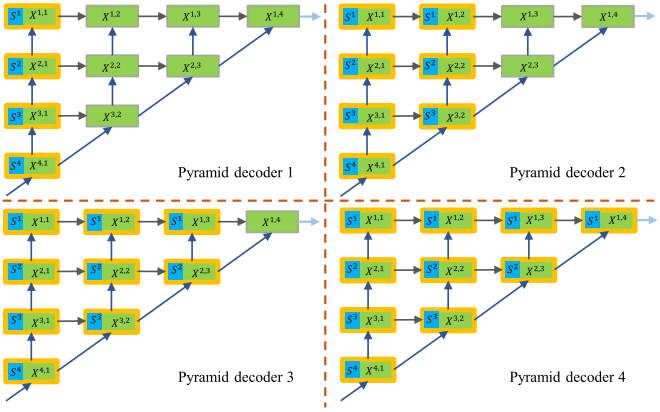


Fig. 11. Pyramid decoder modules with different layer numbers.

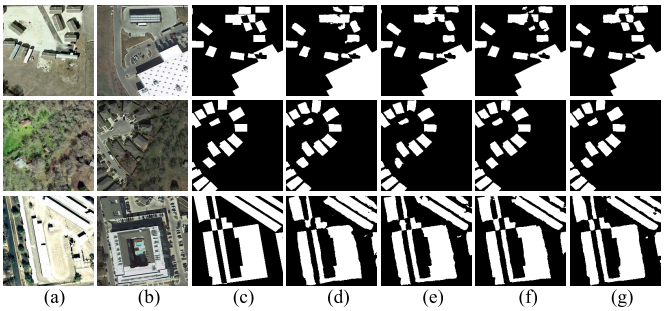


Fig. 12. Experimental results on the LEVIR-CD dataset. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. (d) Fig. 11 (top left). (e) Fig. 11 (top right). (f) Fig. 11 (bottom left). (g) Fig. 11 (bottom right) (Ours).

C. Discussion on the Model Performance on Changed Objects Sizes

As described in the previous parts, the TCD-Net is designed for improving the ability of the network for capturing

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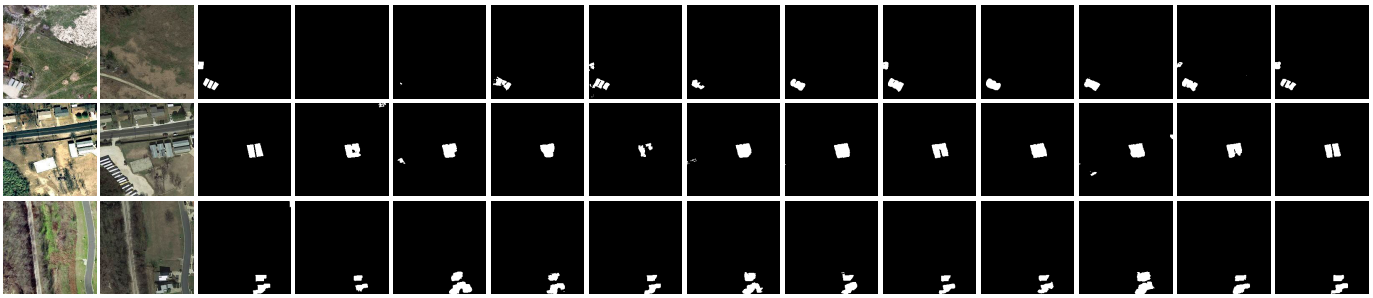


Fig. 13. Comparison of small-sized changed objects on the LEVIR-CD dataset. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. (d) FC-EF. (e) FC-Siam-Di. (f) FC-Siam-Conc. (g) FCN-PP. (h) FDCNN. (i) DFISN. (j) SRCD-Net. (k) Trans-CD. (l) DSAMNet. (m) ChangeFormer. (n) Ours.

TABLE VIII

QUANTITATIVE EVALUATION RESULTS FOR THE ES OF THE PROPOSED NETWORK AND PYRAMID DECODER MODULES ON THE LEVIR-CD DATASET. HIGHER VALUES OF IOU, $F1$, AND DIP INDICATE GOOD CD PERFORMANCE. THE BEST VALUES ARE IN BOLD, AND ALL THE VALUES ARE IN %

Configuration	IoU	$F1$	DIP
ES + the top left of Fig. 11	83.93	90.86	89.85
ES + the top right of Fig. 11	84.12	91.11	90.03
ES + the bottom left of Fig. 11	85.03	91.66	90.64
ES + the bottom right of Fig. 11	**86.45**	**92.53**	**91.57**

TABLE IX

QUANTITATIVE EVALUATION RESULTS FOR DIFFERENT SIZES OF CHANGED OBJECTS ON THE LEVIR-CD DATASET. HIGHER VALUES OF IOU, $F1$, AND DIP INDICATE GOOD CD PERFORMANCE. THE BEST VALUES ARE IN BOLD, AND ALL THE VALUES ARE IN %

Method	Small-sized, area<32×32			Medium-sized, 32×32 <area<96×96		
	IoU	$F1$	DIP	IoU	$F1$	DIP
FC-EF [10]	54.64	70.31	69.33	75.18	85.56	84.16
FC-Siam-Di [10]	59.80	72.96	72.36	74.91	85.38	84.00
FC-Siam-Conc [10]	54.85	66.08	66.86	71.18	82.39	81.11
FCN-PP [11]	61.16	75.62	74.32	77.52	87.29	85.88
FDCNN [29]	62.63	74.40	74.02	77.32	87.00	85.56
DFISN [53]	70.35	82.53	81.01	80.47	88.89	87.68
SRCD-Net [31]	76.25	86.57	85.07	83.55	90.72	89.64
Trans-CD [51]	69.51	80.71	79.65	80.25	88.83	87.55
DSAMNet [33]	71.76	83.88	82.21	79.66	88.57	87.28
ChangeFormer [54]	73.01	84.58	82.97	81.52	89.41	88.33
Ours	**78.75**	**86.73**	**85.96**	**85.22**	**91.81**	**90.79**

representative changed features and improving the model's position awareness of changed objects. Especially, our method can effectively reach better detections of small changed objects. In this section, we define the size of changed objects by investigating [57], including small-sized changed objects (area $<$ 32 $\times$ 32 pixels) and medium-sized changed objects (32 $\times$ 32 pixels $<$ area $<$ 96 $\times$ 96 pixels). In our experiments, the numbers of "small-sized" and "medium-sized" samples are 158 and 1255, respectively. Meanwhile, we report the performance comparison of different methods on small-sized and medium-sized changed objects. The corresponding visual results and quantitative results are shown in Figs. 13 and 14 and Table IX.

From Fig. 13(d)–(f), it can be seen that the first three comparative methods cause more missed detections and show weak detection ability for small changed objects due to the employment of shallow networks. It can also be observed that FDCNN, DFISN, Trans-CD, and DSAMNet roughly detect the regions of small changed objects, resulting in blurred object boundaries as shown in Fig. 13(h), (i), (k), and (l). In contrast, FCN-PP improves the boundary accuracy of small changed objects due to the use of the multiscale feature pyramid module as shown in Fig. 13(g). Moreover, both SRCD-Net and ChangeFormer achieve satisfactory CD results on small changed objects as shown in Fig. 13(j) and (m). Obviously, compared to comparative methods, the proposed method detects small changed objects with clearer boundaries by utilizing the JM-DFE guiding blocks as shown in Fig. 13(d)–(n). Similarly, in Fig. 14(d)–(n), the FST models the long-range dependency among multiscale changed objects by using a full-scale connection strategy, which achieves better detections on changed objects of different scales. In conclusion, the proposed method demonstrates superior detection accuracy and network performance for different CD tasks.

D. Discussion on the Model Complexity

In this section, we further compare the model complexity of comparative methods and the proposed network by presenting another four metrics, including floating point operations (FLOPs), parameters (Params.), model size (Model), MF1 and MDIP, where MF1 denotes the average value of $F1$ on three datasets and MDIP denotes the average value of DIP on three datasets. The quantitative results are reported in Table X. More specifically, FCN-PP corresponds to a larger model size, since it utilizes the Gaussian pyramid module. Compared to FCN-PP, FDCNN has a similar MF1, and it is worth noting that FDCNN requires a small number of parameters and model size due to a double-channel network with a manner of sharing weights. In addition, DFISN improves the MF1 value but increases the model complexity due to the use of deep supervision. For Trans-CD, a modified ViT module is added to U-Net, which results in a slightly larger model size than U-Net

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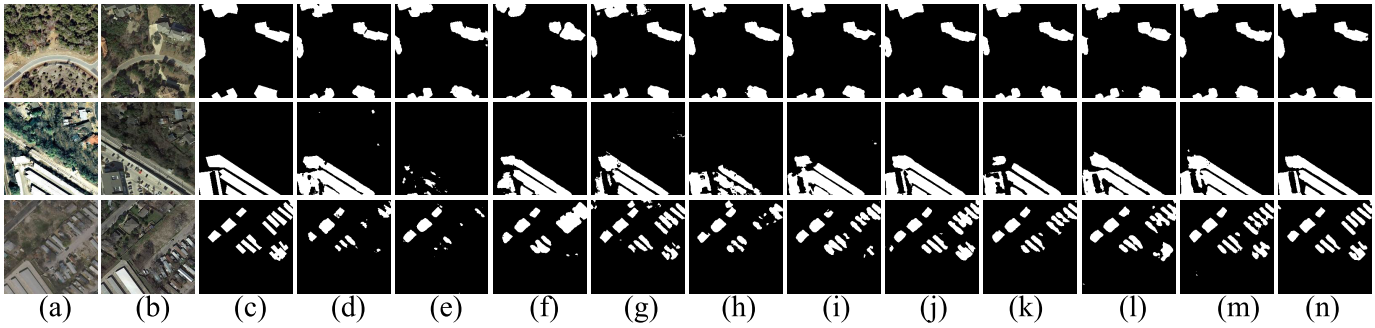


Fig. 14. Comparison of medium-sized changed objects on the LEVIR-CD dataset. (a) Pretemporal images. (b) Posttemporal images. (c) Ground truths. (d) FC-EF. (e) FC-Siam-Di. (f) FC-Siam-Conc. (g) FCN-PP. (h) FDCNN. (i) DFISN. (j) SRCD-Net. (k) Trans-CD. (l) DSAMNet. (m) ChangeFormer. (n) Ours.

TABLE X
QUANTITATIVE COMPARISON OF THE COMPARATIVE METHODS AND OUR NETWORK ON MODEL COMPLEXITY AND ACCURACY, THE BEST VALUES ARE IN BOLD

Methods	FLOPs	Params.	Model	$MF1(\%)$	$MDIP(\%)$
FC-EF[10]	**2.63G**	**0.85M**	**3.34M**	80.72	79.58
FC-Siam-Di[10]	3.47G	0.85M	3.34M	82.35	81.05
FC-Siam-Conc[10]	34.07G	1.07M	4.09M	79.47	78.39
FCN-PP[11]	34.65G	28.13M	107.39M	83.65	82.39
FDCNN[29]	32.40G	1.86M	7.09M	83.45	82.13
DFISN[53]	112.15G	43.50M	166.92M	85.65	84.23
SRCD-Net[31]	27.42G	16.19M	64.86M	86.63	85.36
Trans-CD[51]	39.25G	16.37M	57.27M	84.67	83.32
DSAMNet[33]	57.69G	16.95M	64.75M	85.62	84.33
ChangeFormer[54]	155.13G	41.02M	156.61M	86.31	85.01
Ours	56.64G	11.13M	43.54M	**88.11**	**86.85**

but has lower model complexity than SRCD-Net. Moreover, SRCD-Net and DSAMNet have similar model complexity, and SRCD-Net presents the best value of MF1 among these comparative methods due to the stacked attention module. Moreover, ChangeFormer corresponds to a larger model size since it adopts the Siamese ViT structure.

To summarize, compared to these comparative methods, the proposed structure outperforms the best comparative method by 1.48% and 1.49% on MF1 and MDIP, respectively, and requires much fewer computations. Although the FLOPs, parameters and model size of the proposed method are not the best in Table X, they are moderate compared to the comparative methods and smaller than Transformer-based comparative methods. Besides, the proposed method reaches the best values on MF1 and MDIP metrics, which further demonstrates its superiority in terms of model performance and accuracy. These exceptional results highlight the effectiveness of the proposed method in capturing changed features and identifying changed regions accurately, making it an attractive CD approach in the field of remote sensing. Overall, the proposed method makes a good trade-off between detection accuracy and model complexity, and thus shows clear superiority in CD tasks.

VI. CONCLUSION

In this article, a TCD-Net with JM-DFE and FST is proposed for remote sensing images. The TCD-Net aims to improve the ability of the network for capturing effective changed features and achieve better CD performance. To achieve this, we propose the JM-DFE guiding block based on a joint MFCA module, which can improve the position awareness of truly changed regions. We also present the FST module as the third branch and a pyramid decoder module to produce more compact changed information of multiscale objects. The extensive experimental results on three public datasets reveal that the proposed TCD-Net can extract changed features effectively and show better performance in CD tasks. Moreover, the ablation study and discussion analysis further show that the proposed TCD-Net offers improvements for CD tasks.

REFERENCES

[1] B. L. Keeler et al., "Social-ecological and technological factors moderate the value of urban nature," *Nature Sustainability*, vol. 2, no. 1, pp. 29–38, Jan. 2019.

[2] P.-L. St-Charles, G.-A. Bilodeau, and R. Bergevin, "SuBSENSE: A universal change detection method with local adaptive sensitivity," *IEEE Trans. Image Process.*, vol. 24, no. 1, pp. 359–373, Jan. 2015.

[3] Z. Lv, H. Huang, L. Gao, J. A. Benediktsson, M. Zhao, and C. Shi, "Simple multiscale UNet for change detection with heterogeneous remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[4] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[5] D. Xue, T. Lei, X. Jia, X. Wang, T. Chen, and A. K. Nandi, "Unsupervised change detection using multiscale and multiresolution Gaussian-mixture-model guided by saliency enhancement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1796–1809, 2021.

[6] Q. Zhu et al., "Land-use/land-cover change detection based on a Siamese global learning framework for high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 63–78, Feb. 2022.

[7] T. Lei, D. Xue, Z. Lv, S. Li, Y. Zhang, and A. K. Nandi, "Unsupervised change detection using fast fuzzy clustering for landslide mapping from very high-resolution images," *Remote Sens.*, vol. 10, no. 9, p. 1381, Aug. 2018.

[8] R. Touati, M. Mignotte, and M. Dahmane, "Multimodal change detection in remote sensing images using an unsupervised pixel pairwise-based Markov random field model," *IEEE Trans. Image Process.*, vol. 29, pp. 757–767, 2020.

[9] Z. Lv et al., "Land cover change detection with heterogeneous remote sensing images: Review, progress, and perspective," *Proc. IEEE*, vol. 110, no. 12, pp. 1976–1991, Dec. 2022.

[10] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[11] T. Lei, Y. Zhang, Z. Lv, S. Li, S. Liu, and A. K. Nandi, "Landslide inventory mapping from bitemporal images using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 982–986, Jun. 2019.

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[12] T. Lei et al., "Difference enhancement and spatial–spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4507013.

[13] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[14] J. Hu, L. Shen, and G. Sun, "Squeeze-and-Excitation Networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 7132–7141.

[15] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.

[16] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[17] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[18] M. Gong, J. Zhao, J. Liu, Q. Miao, and L. Jiao, "Change detection in synthetic aperture radar images based on deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 125–138, Jan. 2016.

[19] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019.

[20] O. Akcay, A. C. Kinaci, E. O. Avsar, and U. Aydar, "Semantic segmentation of high-resolution airborne images with dual-stream DeepLabV3+," *ISPRS Int. J. Geo-Inf.*, vol. 11, no. 1, p. 23, Dec. 2021.

[21] T. Lei, D. Xue, H. Ning, S. Yang, Z. Lv, and A. K. Nandi, "Local and global feature learning with kernel scale-adaptive attention network for VHR remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7308–7322, 2022.

[22] C. Wu, B. Du, and L. Zhang, "Fully convolutional change detection framework with generative adversarial network for unsupervised, weakly supervised and regional supervised change detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 9774–9788, Aug. 2023.

[23] H. Jiang, X. Hu, K. Li, J. Zhang, J. Gong, and M. Zhang, "PGA-SiamNet: Pyramid feature-based attention-guided Siamese network for remote sensing orthoimagery building change detection," *Remote Sens.*, vol. 12, no. 3, p. 484, Feb. 2020.

[24] P. Zhang, M. Gong, H. Zhang, J. Liu, and Y. Ban, "Unsupervised difference representation learning for detecting multiple types of changes in multitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2277–2289, Apr. 2019.

[25] M. D. Li et al., "Siamese neural networks for continuous disease severity evaluation and change detection in medical imaging," *npj Digit. Med.*, vol. 3, no. 1, p. 48, Mar. 2020.

[26] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[27] D. Lu, S. Cheng, L. Wang, and S. Song, "Multi-scale feature progressive fusion network for remote sensing image change detection," *Sci. Rep.*, vol. 12, no. 1, p. 11968, Jul. 2022.

[28] P. He et al., "MANet: Multi-scale aware-relation network for semantic segmentation in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624615.

[29] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[30] Z. Duan, S. Wang, H. Di, and J. Deng, "Distillation remote sensing object counting via multi-scale context feature aggregation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613012.

[31] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4403718.

[32] L. Wang, L. Wang, Q. Wang, and P. M. Atkinson, "SSA-SiamNet: Spectral–spatial-wise attention-based Siamese network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510018.

[33] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[34] M. Papadomanolaki, S. Verma, M. Vakalopoulou, S. Gupta, and K. Karantzalos, "Detecting urban changes with recurrent neural networks from multitemporal Sentinel-2 data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jun. 2019, pp. 214–217.

[35] S. Sun, L. Mu, L. Wang, and P. Liu, "L-UNet: An LSTM network for remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[36] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[37] I. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, pp. 5998–6008, 2017.

[38] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[39] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[40] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[41] W. Zhang, M. Zhou, C. Ji, X. Sui, and J. Bai, "Cross-frame transformer-based spatio-temporal video super-resolution," *IEEE Trans. Broadcast.*, vol. 68, no. 2, pp. 359–369, Jun. 2022.

[42] N. Carion et al., "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 213–229.

[43] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[44] Z. Liu et al., "Swin Transformer: Hierarchical vision Transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[45] Y. Zhou, S. Chen, J. Zhao, R. Yao, Y. Xue, and A. E. Saddik, "CLT-Det: Correlation learning based on transformer for detecting dense objects in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4708915.

[46] Z. Wang, J. Guo, C. Zhang, and B. Wang, "Multiscale feature enhancement network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5634819.

[47] L. Ding et al., "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," 2021, *arXiv:2106.15754*.

[48] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

[49] P. Lv, W. Wu, Y. Zhong, F. Du, and L. Zhang, "SCViT: A spatial-channel feature preserving vision transformer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409512.

[50] K. Xu, P. Deng, and H. Huang, "Vision transformer: An excellent teacher for guiding small networks in remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618715.

[51] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[52] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213.

[53] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[54] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 207–210.

[55] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[56] A. K. Nandi, "From multiple independent metrics to single performance measure based on objective function," *IEEE Access*, vol. 11, pp. 3899–3913, 2023.

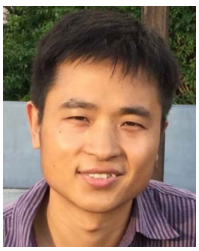
[57] J. Li, X. Liang, Y. Wei, T. Xu, J. Feng, and S. Yan, "Perceptual generative adversarial networks for small object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2017, pp. 1222–1230.

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